

LOGISTIC REGRESSION PROJECT REPORT

Project: Titanic Survival Prediction using Kaggle Data

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Objective	Dataset	Model
Predict Survived: 0 = Did not survive, 1 = Survived.	Titanic - Machine Learning from Disaster from Kaggle; file: train.csv (891 passengers).	Median imputation + StandardScaler + balanced Logistic Regression.

Dataset and Workflow

Inputs: Pclass, Sex, Age, SibSp, Parch, Fare, Embarked, plus engineered FamilySize and IsAlone. PassengerId, Name, Ticket and Cabin are dropped.

Workflow: load Kaggle CSV → inspect data → EDA → encode categories and cap Fare outliers → train-test split → median imputation → scaling → train Logistic Regression → evaluate → interpret coefficients.

Evaluation Metrics

Metric	Result	Meaning
Accuracy	0.788	Overall correct predictions
Precision	0.707	Reliability of positive predictions
Recall	0.768	Actual positive cases detected
F1-score	0.736	Balance of precision and recall
ROC-AUC	0.846	Overall class-separation ability

Confusion Matrix

	Predicted 0	Predicted 1
Actual 0	TN = 88	FP = 22
Actual 1	FN = 16	TP = 53

Key Findings

The model obtains a ROC-AUC of about 0.846 and detects approximately 76.8% of the survivors in the fixed test split. Being male (Sex_male, coefficient -1.21) and a lower passenger class (Pclass, -0.67) have the strongest negative coefficients, while Fare has the strongest positive coefficient (0.35). Recall matters here because false negatives represent survivors the model missed.